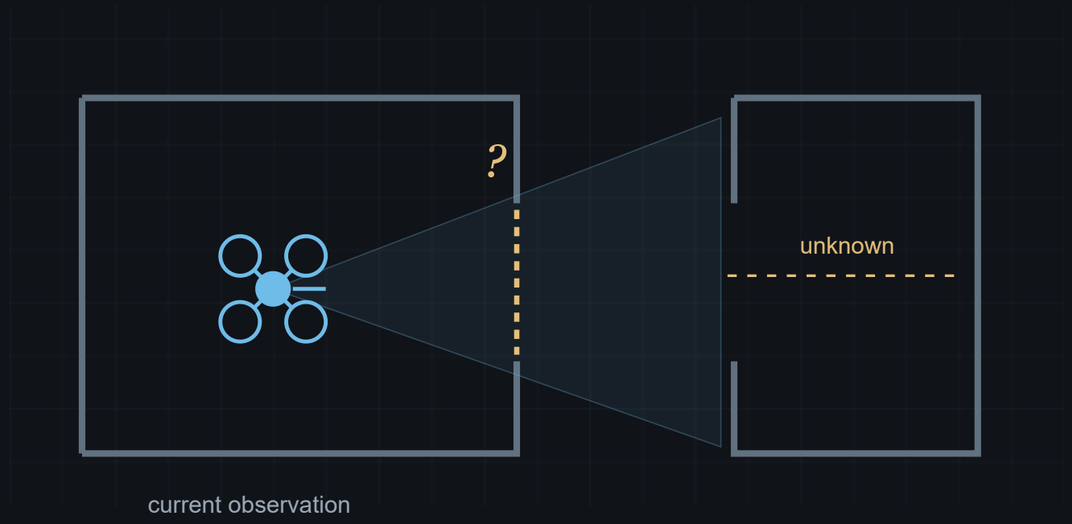
The course is visible.
The next action is uncertain.

Indoor navigation, incomplete observations, and a mission beyond the field of view.



What should happen next?

A passage candidate

What lies beyond it?

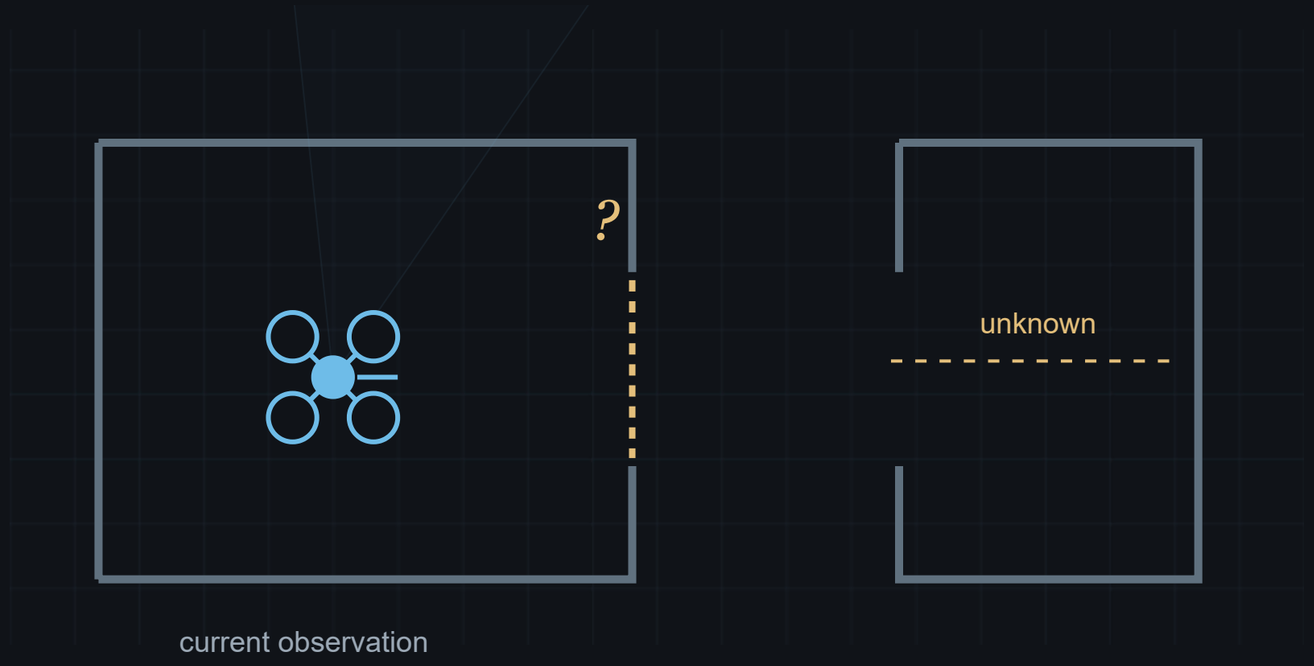
Observe before committing.

geometry + memory + intent

↳ Each camera frame reveals only part of the task.

Build what the drone *remembers*.

Persistent identity. Evidence. Uncertainty. A reason to observe again.



Memory ≠ a fresh view

opening_0003

same entity, new viewpoint

last seen → aging

identity persists · certainty does not

↳ Prioritize a useful world representation before increasing policy complexity.

Illustrative belief update

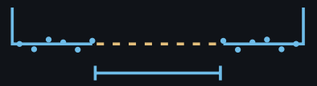
Combine ideas at the interfaces.

Research gave us the structure; our constraints shaped the implementation.

RESEARCH → ADAPTATION

Hydra

incremental spatial hierarchy

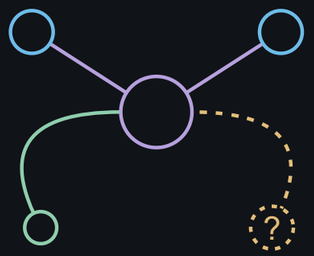


Kimera

metric + semantic structure



Compact WorldState



geometry + identity + evidence

CSIRO: recovery in partial observation

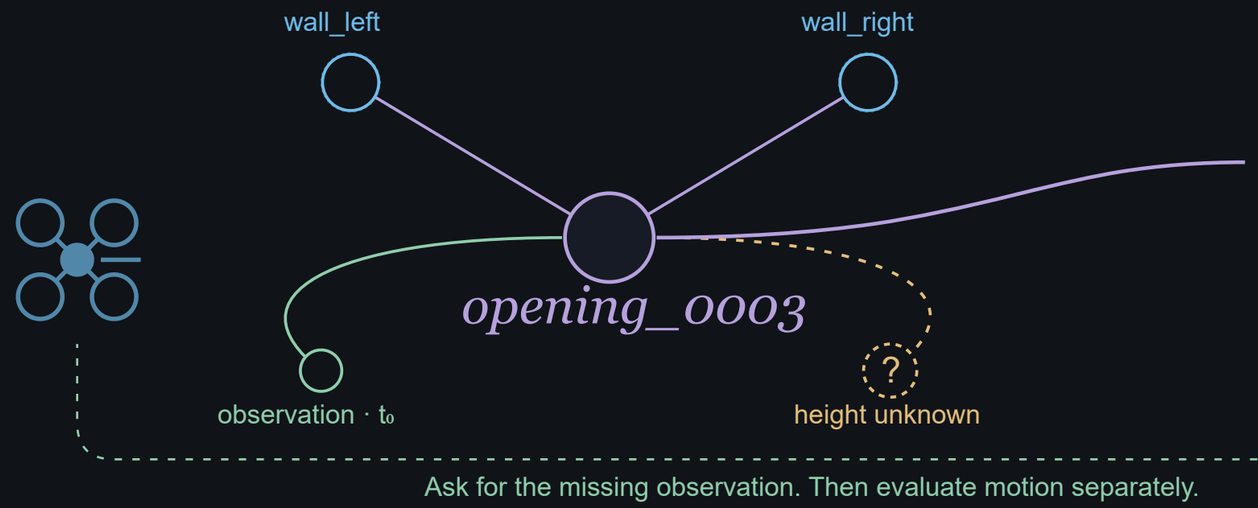
ModalAI: separate perception, planning and autopilot interfaces

↳ A compact adaptation for our task and compute budget.

From seeing to *understanding*.

Observations become geometry, then meaning, then a grounded decision.

01 OBSERVATIONS | 02 GEOMETRY | 03 SEMANTIC GRAPH | 04 REASONING



GOAL PROPOSAL

observe_more

target: opening_0003

because: height is unknown

Grounded proposal ≠ permission to fly

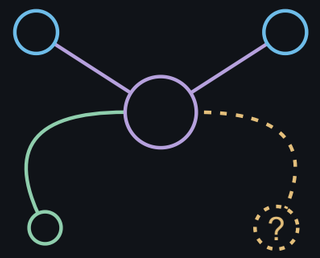
Ask for the missing observation. Then evaluate motion separately.

↳ Missing evidence becomes a reason to look again.

Illustrative entity · offline contract

Sometimes the next action is *to look*.

A bounded reasoning layer turns missing information into an explicit goal.



opening_0003

height remains unknown

Bound the context

entity + evidence

frame + time + missing facts

no invented connectivity

Choose the next question

observe_more

target: opening_0003

because: height is unknown

Grounding + freshness validated

Motion still needs its own feasibility check.

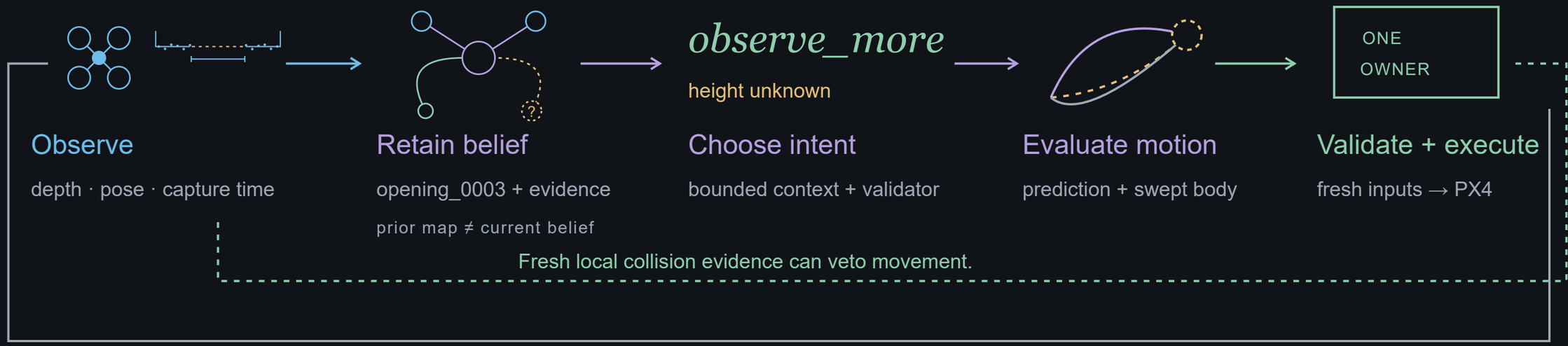
↳ “Observe this opening again” is a useful decision.

Non-executable proposal

Each layer answers a different question.

Our target architecture separates memory, intent, motion and execution.

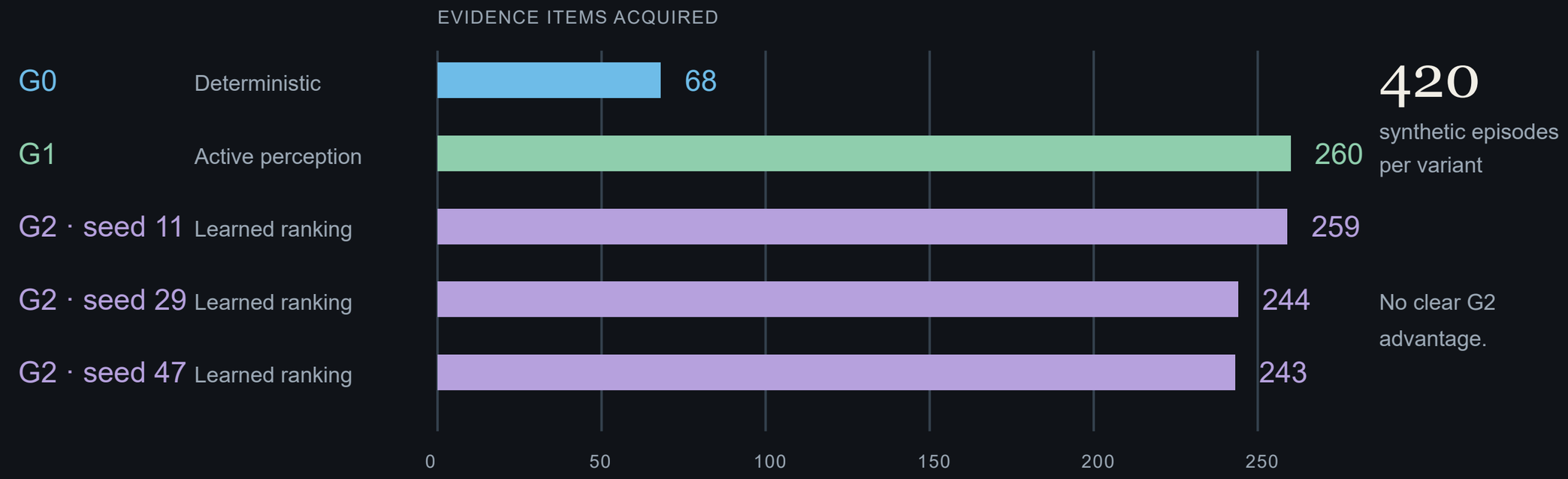
THE SAME EVIDENCE, THROUGH DIFFERENT RESPONSIBILITIES



↳ Persistent belief and fresh collision evidence have different jobs.

Complexity has to earn its place.

Active perception earned priority in a synthetic evidence-acquisition task.



420
synthetic episodes per variant

No clear G2 advantage.

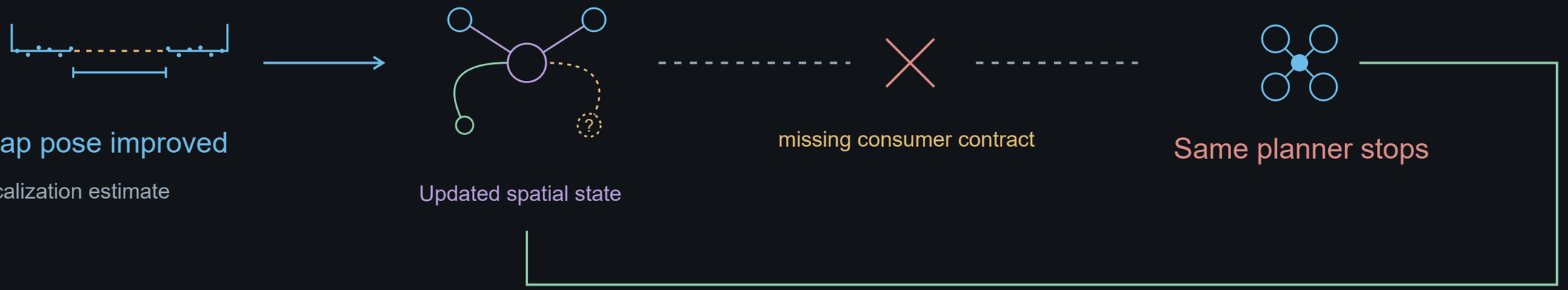
↳ Active perception improved this task; learned ranking showed no clear advantage.

Synthetic benchmark · not flight success

A better map was not yet a better decision.

An integration lesson: better estimates must reach the decisions they are meant to change.

HISTORICAL P2 FIXTURE

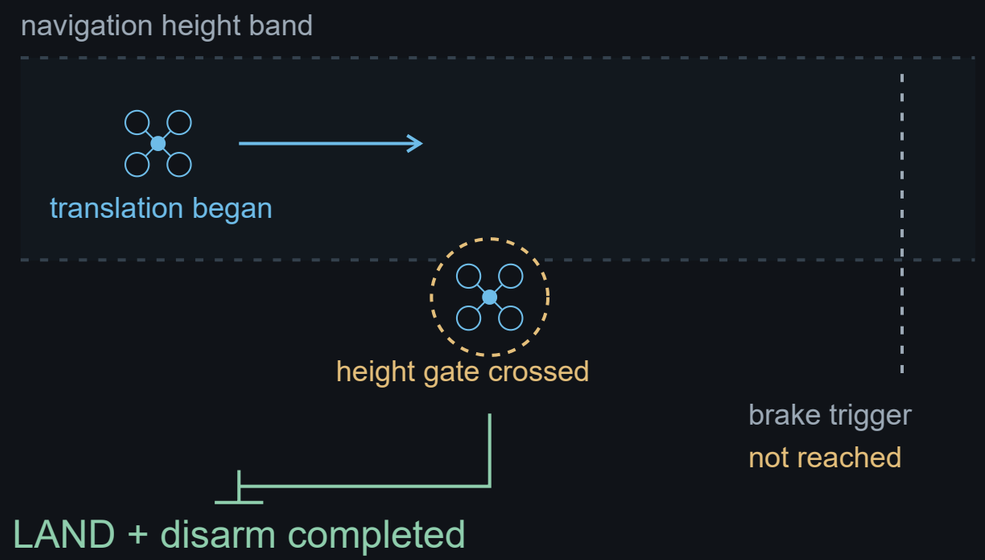


↳ We shifted toward explicit consumer contracts and bounded asynchronous registration.

The assumptions failed between layers.

Native experiments exposed altitude, frame, timing and scheduling contracts.

C1 · THE BRAKING MEASUREMENT FAILED



Schematic states and event order · not a measured trajectory

S0 · FOUR DISARMED ATTEMPTS



01 Frame origins

bind the same physical position



02 Clock alignment

preserve capture-time identity



03–04 Readiness + scheduling

bound the work on the owner path

No course-flight success claimed.

↳ Preserved failures became specific engineering work.

What we built. What we still need to prove.

Evidence is specific to a component, an experiment and an environment.

EVIDENCE IS LOCAL TO THE CLAIM



Offline components

entities · reasoning · mapping

synthetic comparison evidence



Native calibration

C0: completed

C1: recovered; measurement failed



Mission-level benefit

hybrid course + matched trials

hardware qualification remains

Separate baseline: Classical S1 has one successful Gazebo course run.

↳ The hybrid architecture has not yet demonstrated full-course native success in the fetched record.

From observations to *justified actions*.

The next milestone is a measurable mission benefit from each layer.



01 Reliable native segment

frames · timing · height control



02 Matched comparison

same map, start and scoring



03 Layer ablations

memory → topology → semantics

Observations ⟶ *Grounded belief* ⟶ *Justified actions*

Learned extensions only when the evidence warrants them.

↳ Our contribution is a testable path from observations to justified actions.

N1 · N6 · N7 · N8

Planned evaluation